

11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

- a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(PRICE)$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?
- b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.
- c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.
- d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

a.

$$\ln(\hat{price}) = \beta_0 + \beta_1 \text{MON} + \beta_2 \text{TUE} + \beta_3 \text{WED} + \beta_4 \text{THU} + \beta_5 \text{STORMY} + \beta_6 \text{MIXED}$$

Call:
lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed,
data = fultonfish)

Residuals:

	Min	1Q	Median	3Q	Max
	-0.92281	-0.18819	0.00813	0.22404	0.68381

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.35957	0.07915	-4.543	1.50e-05 ***
mon	-0.10825	0.10339	-1.047	0.29753
tue	-0.06609	0.10103	-0.654	0.51446
wed	-0.04927	0.10377	-0.475	0.63591
thu	0.03935	0.10071	0.391	0.69681
stormy	0.44634	0.07921	5.635	1.51e-07 ***
mixed	0.23688	0.07851	3.017	0.00321 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3413 on 104 degrees of freedom
Multiple R-squared: 0.245, Adjusted R-squared: 0.2014
F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05

i P-value = 0.00321 < 0.05

∴ significant

表示天氣狀況對價格

有顯著影響

Linear hypothesis test:

stormy = 0

mixed = 0

Model 1: restricted model

Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$H_0: \text{stormy} = 0, \text{mixed} = 0$

$F^* = 16.14 > 10$, reject H_0

有證據支持 stormy 和 mixed 可以解釋 price, 可以做輸入工具變數

b. Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
tue + wed + thu + stormy + mixed, data = fultonfish)

Residuals:

	Min	1Q	Median	3Q	Max
	-2.1327	-0.4320	0.0654	0.5133	1.1798

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.54023	0.15661	54.533	< 2e-16 ***
lprice	-0.93014	0.35340	-2.632	0.00977 **
mon	-0.01190	0.20837	-0.057	0.95456
tue	-0.52583	0.20230	-2.599	0.01068 *
wed	-0.56262	0.20696	-2.719	0.00767 **
thu	0.09987	0.20285	0.492	0.62350

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared: 0.185, Adjusted R-squared: 0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317

TABLE 11.5 2SLS Estimates for Fish Demand

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	8.5059	0.1662	51.1890	0.0000
ln(PRICE)	-1.1194	0.4286	-2.6115	0.0103
MON	-0.0254	0.2148	-0.1183	0.9061
TUE	-0.5308	0.2080	-2.5518	0.0122
WED	-0.5664	0.2128	-2.6620	0.0090
THU	0.1093	0.2088	0.5233	0.6018

$$\ln(\text{Quantity}) = 8.5402 - 0.9301 \ln(\text{Price}) - 0.0119 \text{MON} - 0.5258 \text{TUE} - 0.5626 \text{WED} + 0.0999 \text{THU}$$

當增MTRD做IV後, 需求價格彈性的估計值由-1.1194變為-0.9301
但仍符合需求法則. 在 $\alpha=0.05$ 下, 仍顯著.

SE由0.4286變為0.3534. SE下降, 估計效率提升, 表示加入了一個有效的IV, 使得對需求彈性的估計更精確.

c.

Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
tue + wed + thu + stormy + mixed, data = fultonfish)

Residuals:

	Min	1Q	Median	3Q	Max
	-2.1327	-0.4320	0.0654	0.5133	1.1798

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.54023	0.15661	54.533	< 2e-16 ***
lprice	-0.93014	0.35340	-2.632	0.00977 **
mon	-0.01190	0.20837	-0.057	0.95456
tue	-0.52583	0.20230	-2.599	0.01068 *
wed	-0.56262	0.20696	-2.719	0.00767 **
thu	0.09987	0.20285	0.492	0.62350

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	2	104	16.140	7.86e-07 ***
Wu-Hausman	1	104	1.489	0.225
Sargan	1	NA	0.772	0.380

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared: 0.185, Adjusted R-squared: 0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317

H_0 : Instruments are valid

$\therefore p\text{-value} = 0.38 > 0.05$

$\therefore$ 無法拒絕 H_0

表示STORMY和MTRD是

有效的工具變數

d.

Linear hypothesis test:

mon = 0

tue = 0

wed = 0

thu = 0

Model 1: restricted model

Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

$H_0: \text{MON} = \text{TUE} = \text{WED} = \text{THU} = 0$ v.s H_1 : At least one non-zero

$\therefore p\text{-value} = 0.6437 > 0.05$

$\therefore$ 無法拒絕 H_0 . 表示在控制天氣因素後, 日期虛假變數對於價格變動沒有顯著的解釋力

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?
- b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?
- c. Estimate the price elasticity of demand “at the means” using the results from (b).
- d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.
- e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?
- f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

a. Demand: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$ (11.11)

Supply: $Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$ (11.12)

$$P_i = \frac{1}{\alpha_2} (\alpha_1 - \alpha_3 PS_i - \alpha_4 DI_i - e_{di})$$

$$= \frac{-\alpha_1}{\alpha_2} + \frac{1}{\alpha_2} \alpha_1 - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di}$$

① 預期: Q_i 係數 < 0 ; 需求法則, 價格又需求是反向變動

$$\alpha_2 < 0 \Rightarrow \frac{1}{\alpha_2} < 0$$

② PS_i 係數 > 0 ; 替代品變貴, 需求增加
替代品價格

$$\alpha_2 < 0, \alpha_3 > 0 \Rightarrow -\frac{\alpha_3}{\alpha_2} > 0$$

③ DI_i 係數 > 0 ; 假設 normal good, $DI \uparrow$, 需求 $\uparrow$
可支配收入

$$\alpha_2 < 0, \alpha_4 > 0 \Rightarrow -\frac{\alpha_4}{\alpha_2} > 0$$

$$P_i = \frac{1}{\beta_2} (\alpha_1 - \beta_1 - \beta_3 PF_i - e_{si}) = -\frac{\beta_1}{\beta_2} + \frac{1}{\beta_2} \alpha_1 - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si}$$

① 預期: Q_i 係數 > 0 ; 供給法則

$$\beta_2 > 0 \Rightarrow \frac{1}{\beta_2} > 0$$

② PF_i 係數 > 0
生產要素價格
 $\beta_2 > 0, \beta_3 < 0 \Rightarrow -\frac{\beta_3}{\beta_2} > 0$

b.

Call:
ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)

→ Demand

Residuals:

	Min	1Q	Median	3Q	Max
	-39.661	-6.781	2.410	8.320	20.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.428	13.592	-0.841	0.40810
q	-2.671	1.175	-2.273	0.03154 *
ps	3.461	1.116	3.103	0.00458 **
di	13.390	2.747	4.875	4.68e-05 ***

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	1	26	17.48	0.000291 ***
Wu-Hausman	1	25	120.03	4.92e-11 ***
Sargan	0	NA	NA	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567, Adjusted R-squared: 0.5056
Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06

符号符合預期 ($Q < 0$, $PS > 0$, $DI > 0$)

p-value 皆小於 0.05 ⇒ 皆顯著

Call:
ivreg(formula = p ~ q + pf | ps + di + pf, data = truffles)

→ Supply

Residuals:

	Min	1Q	Median	3Q	Max
	-9.7983	-2.3440	-0.6281	2.4350	11.1600

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-58.7982	5.8592	-10.04	1.32e-10 ***
q	2.9367	0.2158	13.61	1.32e-13 ***
pf	2.9585	0.1560	18.97	< 2e-16 ***

Diagnostic tests:

	df1	df2	statistic	p-value
Weak instruments	2	26	28.934	2.44e-07 ***
Wu-Hausman	1	26	7.051	0.0134 *
Sargan	1	NA	1.544	0.2140

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486, Adjusted R-squared: 0.9448
Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16

符号符合預期 ($Q > 0$, $PF > 0$)

p value 皆小於 0.05 ⇒ 皆顯著

c.

```

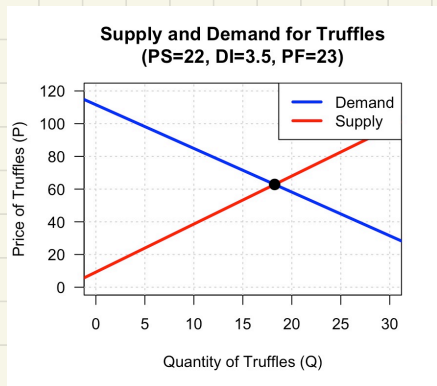
> cat(sprintf("1. 樣本平均價格 (P_bar) : %.4f\n", mean_p))
1. 樣本平均價格 (P_bar) : 62.7240
> cat(sprintf("2. 樣本平均數量 (Q_bar) : %.4f\n", mean_q))
2. 樣本平均數量 (Q_bar) : 18.4583
> cat(sprintf("3. 反需求迴歸斜率 (dP/dQ): %.4f\n", gamma_2))
3. 反需求迴歸斜率 (dP/dQ): -2.6705
> cat(sprintf("4. 邊際效果 (dQ/dP) : %.4f\n", 1 / gamma_2))
4. 邊際效果 (dQ/dP) : -0.3745
> cat(sprintf("-----\n"))
-----
> cat(sprintf("► 需求價格彈性 (E_d) : %.4f\n", elasticity))
► 需求價格彈性 (E_d) : -1.2725

```

price elasticity of demand = -1.2725

因為絕對值 > 1, 所以在平均時, 需求是具有彈性的.

d.



```

> cat(sprintf("固定後需求線: P = %.4f %.4f Q\n", int_demand, slope_demand))
固定後需求線: P = 111.5801 -2.6705 Q
> cat(sprintf("固定後供給線: P = %.4f + %.4f Q\n", int_supply, slope_supply))
固定後供給線: P = 9.2470 + 2.9367 Q

```

e. === 根據 Table 11.2 係數計算的結果 ===
> cat(sprintf("預測均衡數量 Q* = %.4f\n", Q_pred_table))
預測均衡數量 Q* = 18.2601
> cat(sprintf("預測均衡價格 P* = %.4f\n", P_pred_table))
預測均衡價格 P* = 61.5436

=== 實際資料跑出的縮減式係數 (驗證 Table 11.2) ===

```
> cat("依變數 Q 的係數:\n")  
依變數 Q 的係數:  
> print(round(coef(reduced_form_Q), 4))  
(Intercept)      ps      di      pf  
    7.8951    0.6564    2.1672   -0.5070  
> cat("\n依變數 P 的係數:\n")
```

```
依變數 P 的係數:  
> print(round(coef(reduced_form_P), 4))  
(Intercept)      ps      di      pf  
   -32.5124    1.7081    7.6025    1.3539
```

=== 使用資料模型預測的結果 ===
> cat(sprintf("預測均衡數量 Q* = %.4f\n", Q_pred_data))
預測均衡數量 Q* = 18.2604
> cat(sprintf("預測均衡價格 P* = %.4f\n", P_pred_data))
預測均衡價格 P* = 62.8154

兩看得到的結果相近, 表示 2SLS 估計結構式的預測,
和用 OLS 估計縮減式的預測是一致的。

f.

```
Call:
lm(formula = p ~ q + ps + di, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-25.0753  -2.7742  -0.4097   4.7079  17.4979

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.6195     9.0872  -1.499   0.1460
q             0.1512     0.4988   0.303   0.7642
ps            1.3607     0.5940   2.291   0.0303 *
di            12.3582     1.8254   6.770 3.48e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared:  0.8013,    Adjusted R-squared:  0.7784
F-statistic: 34.95 on 3 and 26 DF,  p-value: 2.842e-09
```

Q的符号不符合預期 ($Q < 0$)

$Q > 0 \Rightarrow$ 違反需求法則

$p\text{-value} > 0.05 \Rightarrow$ 不顯著

```
> print(summary(supply_ols))

Call:
lm(formula = p ~ q + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-8.4721 -3.3287   0.1861   2.0785  10.7513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -52.8763     5.0238  -10.53 4.68e-11 ***
q             2.6613     0.1712   15.54 5.42e-15 ***
pf            2.9217     0.1482   19.71 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared:  0.9531,    Adjusted R-squared:  0.9496
F-statistic: 274.4 on 2 and 27 DF,  p-value: < 2.2e-16
```

符号符合預期 ($Q > 0, PF > 0$)

$p\text{ value}$ 皆小於 $0.05 \Rightarrow$ 皆顯著

```
【需求方程式中 Q 的斜率】
> cat(sprintf(" - 錯誤的 OLS 估計 : %8.4f (產生聯立偏誤)\n", coef(demand_ols)["q"]))
- 錯誤的 OLS 估計 :    0.1512 (產生聯立偏誤)
> cat(sprintf(" - 正確的 2SLS 估計 : %8.4f (符合需求法則)\n\n", coef(demand_2sls)["q"]))
- 正確的 2SLS 估計:  -2.6705 (符合需求法則)

> cat(sprintf("【供給方程式中 Q 的斜率】 \n"))
【供給方程式中 Q 的斜率】
> cat(sprintf(" - 偏誤的 OLS 估計 : %8.4f \n", coef(supply_ols)["q"]))
- 偏誤的 OLS 估計 :    2.6613
> cat(sprintf(" - 正確的 2SLS 估計 : %8.4f \n", coef(supply_2sls)["q"]))
- 正確的 2SLS 估計:    2.9367
```

如果忽略內生性, 直接用OLS估計, 會產生聯立偏誤.

因此需使用 2SLS 來獲得不偏且一致的結果